Table 1: Average long-term forecasting results on the benchmark datasets. We followed the same setting in DMMV [Shen et al., 2025]. Lower MSE and MAE indicate better performance. We set the forecasting horizons $H \in \{24, 36, 48, 60\}$ for Illness and $\{96, 192, 336, 720\}$ for the others. **Red**: the best, Blue: the second best. More results are in Appendix B.

View	**Language**						**Multi-Modal**				**Visual**		**Numerical**									
Methods	RDTU (**Ours**)		GPT4TS [2023]		Time-LLM [2024]		DMMV-A [2025]		Time-VLM [2025]		VisionTS [2024]		PatchTST [2023]		CycleNet [2024]		TimesNet [2023]		DLinear [2023]		FEDformer [2022]	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	0.399	**0.413**	0.418	0.421	0.418	0.432	**0.395**	0.414	0.405	0.420	0.407	0.419	0.413	0.431	0.415	0.426	0.458	0.450	0.423	0.430	0.440	0.460
ETTh2	0.337	**0.377**	0.354	0.389	0.361	0.396	0.337	0.388	0.341	0.391	0.351	0.386	**0.330**	0.379	0.355	0.398	0.414	0.427	0.431	0.447	0.437	0.449
ETTm1	**0.340**	**0.367**	0.363	0.378	0.356	0.377	**0.340**	0.371	0.351	0.376	0.344	0.373	0.351	0.381	0.355	0.379	0.400	0.406	0.357	0.379	0.448	0.452
ETTm2	0.250	**0.309**	0.254	0.311	0.261	0.316	0.256	0.317	**0.248**	0.311	0.267	0.327	0.255	0.315	0.251	**0.309**	0.291	0.333	0.267	0.332	0.305	0.349
Illness	1.559	0.812	1.871	0.852	2.018	0.894	**1.407**	**0.771**	–	–	1.482	0.796	1.443	0.798	2.187	0.992	2.139	0.931	2.169	1.041	2.847	1.144
Electricity	**0.156**	**0.246**	0.170	0.263	0.165	0.259	0.158	0.248	0.172	0.272	0.159	0.250	0.162	0.253	0.158	0.250	0.193	0.295	0.166	0.264	0.214	0.327
Weather	0.221	**0.251**	0.227	0.255	0.244	0.270	**0.217**	0.256	0.224	0.263	0.225	0.258	0.226	0.264	0.242	0.278	0.259	0.287	0.249	0.300	0.309	0.360
Traffic	0.389	**0.255**	0.421	0.274	0.422	0.281	0.389	0.257	0.419	0.304	**0.386**	0.256	0.391	0.264	0.421	0.289	0.620	0.336	0.434	0.295	0.610	0.376
# Wins	**9**		0		0		5		1		1		1		1		0		0		0	

Table 2: Brief short-term time series forecasting results on M4. The forecasting horizons are in [6, 48] and the three rows provided are weighted averaged from all datasets under different sampling intervals. We followed the same setting in Jin et al. [2024].A lower value indicates better performance. **Red**: the best, Blue: the second best. More results are in Appendix B.

	Methods	RDTU (**Ours**)	Time-LLM [2024]	GPT4TS [2023]	TimesNet [2023]	PatchTST [2023]	N-HiTS [2023]	N-BEATS [2020]	ETSformer [2022]	LightTS [2022]	DLinear [2023]	FEDformer [2022]	Stationary [2022]	Autoformer [2021]	Informer [2021]	Reformer [2020]
Average	SMAPE	**11.979**	11.983	12.69	12.88	12.059	12.035	12.25	14.718	13.525	13.639	13.16	12.780	12.909	14.086	18.200
	MASE	1.599	**1.595**	1.808	1.836	1.623	1.625	1.698	2.408	2.111	2.095	1.775	1.756	1.771	2.718	4.223
	OWA	**0.859**	**0.859**	0.94	0.955	0.869	0.869	0.896	1.172	1.051	1.051	0.949	0.930	0.939	1.230	1.775

Table 3: Average few-shot learning results on 10% training data. **Red**: the best, Blue: the second best. More results are in Appendix C.

Methods	RDTU (**Ours**)		Time-LLM [2024]		GPT4TS [2023]		DLinear [2023]		PatchTST [2023]		TimesNet [2023]		FEDformer [2022]		Autoformer [2021]		Stationary [2022]		ETSformer [2022]		LightTS [2022]		Informer [2021]		Reformer [2020]	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	**0.551**	**0.513**	0.556	0.522	0.590	0.525	0.691	0.600	0.633	0.542	0.869	0.628	0.639	0.561	0.702	0.596	0.915	0.639	1.180	0.834	1.375	0.877	1.199	0.809	1.249	0.833
ETTh2	**0.367**	**0.389**	0.370	0.394	0.397	0.421	0.605	0.538	0.415	0.431	0.479	0.465	0.466	0.475	0.488	0.499	0.462	0.455	0.894	0.713	2.655	1.160	3.872	1.513	3.485	1.486
ETTm1	**0.402**	**0.424**	0.404	0.427	0.464	0.441	0.411	0.429	0.501	0.466	0.677	0.537	0.722	0.605	0.802	0.628	0.797	0.578	0.980	0.714	0.971	0.705	1.192	0.821	1.426	0.856
ETTm2	**0.274**	**0.319**	0.277	0.323	0.293	0.335	0.316	0.368	0.296	0.343	0.320	0.353	0.463	0.488	1.342	0.930	0.332	0.366	0.447	0.487	0.987	0.756	3.370	1.440	3.978	1.587
Weather	**0.232**	**0.270**	0.234	0.273	0.238	0.275	0.241	0.283	0.242	0.279	0.279	0.301	0.284	0.324	0.300	0.342	0.318	0.323	0.318	0.360	0.289	0.322	0.597	0.495	0.546	0.469
Electricity	**0.173**	**0.268**	0.175	0.270	0.176	0.269	0.180	0.280	0.180	0.273	0.323	0.392	0.346	0.427	0.431	0.478	0.444	0.480	0.660	0.617	0.441	0.489	1.195	0.891	0.965	0.768
Traffic	**0.426**	**0.303**	0.429	0.306	0.440	0.310	0.447	0.313	0.430	0.305	0.951	0.535	0.663	0.425	0.749	0.446	1.453	0.815	1.914	0.936	1.248	0.684	1.534	0.811	1.551	0.821
# Wins	**14**		0		0		0		0		0		0		0		0		0		0		0		0	

Table 4: Average zero-shot learning results on ETT datasets. A lower value indicates better performance. **Red**: the best, Blue: the second best, *Green*: the third best. Full results are shown in Appendix C.

Methods	RDTU (**Ours**)		Time-LLM [2024]		LLMTime [2023]		GPT4TS [2023]		DLinear [2023]		PatchTST [2023]		TimesNet [2023]		Autoformer [2021]	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
$ETTh1 \rightarrow ETTh2$	**0.344**	**0.379**	0.353	0.387	0.992	0.708	0.406	0.422	0.493	0.488	*0.380*	*0.405*	0.421	0.431	0.582	0.548
$ETTh1 \rightarrow ETTm2$	**0.266**	**0.333**	0.273	0.340	1.867	0.869	0.325	0.363	0.415	0.452	*0.314*	*0.360*	0.327	0.361	0.457	0.483
$ETTh2 \rightarrow ETTh1$	0.480	**0.471**	**0.479**	0.474	1.961	0.981	0.757	0.578	0.703	0.574	*0.565*	*0.513*	0.865	0.621	0.757	0.608
$ETTh2 \rightarrow ETTm2$	**0.268**	**0.329**	0.272	0.341	1.867	0.869	0.335	0.370	0.328	0.386	*0.325*	*0.365*	0.342	0.376	0.366	0.411
$ETTm1 \rightarrow ETTh2$	**0.378**	**0.408**	0.381	0.412	0.992	0.708	*0.433*	0.439	0.464	0.475	0.439	*0.438*	0.457	0.454	0.470	0.479
$ETTm1 \rightarrow ETTm2$	**0.261**	**0.316**	0.268	0.320	1.867	0.869	0.313	0.348	0.335	0.389	*0.296*	*0.334*	0.322	0.354	0.469	0.484
$ETTm2 \rightarrow ETTh2$	**0.351**	**0.395**	0.354	0.400	0.992	0.708	0.435	0.443	0.455	0.471	*0.409*	*0.425*	0.435	0.443	0.423	0.439
$ETTm2 \rightarrow ETTm1$	**0.412**	**0.434**	0.414	0.438	1.933	0.984	0.769	0.567	0.649	0.537	*0.568*	*0.492*	0.769	0.567	0.755	0.591

Table 5: Ablation study of different reward components in the RDFR training stage. Improvements are computed relative to Qwen2.5-Instruct-STIT.

Method	**MSE ↓**	**MAE ↓**
Qwen2.5-Instruct (Zero-shot)	–	–
Qwen2.5-Instruct-STIT	0.403	0.428
+ R_{len}	0.401 (-0.50%)	0.425 (-0.70%)
+ R_{fmt}	0.395 (-1.99%)	0.419 (-2.10%)
+ R_{acc}	0.367 (-8.93%)	0.398 (-7.01%)
+ $R_{len} + R_{fmt}$	0.388 (-3.72%)	0.412 (-3.74%)
RDTU (R_{total})	**0.351 (-12.90%)**	**0.382 (-10.75%)**

Table 6: Training settings and resource usage on different datasets.

Dataset	**Stage**	**Training Size**	**Epochs**	**Time (Hours)**	**Peak VRAM per GPU**
ETTh1	STIT / RDFR	8,545 / 425	15 / 3	$\sim$2.5 / $\sim$0.3	$\sim$24 GB / $\sim$32 GB
ETTh2	STIT / RDFR	8,545 / 425	15 / 3	$\sim$2.5 / $\sim$0.3	$\sim$24 GB / $\sim$32 GB
ETTm1	STIT / RDFR	34,465 / 1,713	5 / 2	$\sim$4.3 / $\sim$0.5	$\sim$24 GB / $\sim$32 GB
ETTm2	STIT / RDFR	34,465 / 1,713	5 / 2	$\sim$4.3 / $\sim$0.5	$\sim$24 GB / $\sim$32 GB
Electricity	STIT / RDFR	18,317 / 911	8 / 3	$\sim$3.7 / $\sim$0.5	$\sim$24 GB / $\sim$32 GB
Traffic	STIT / RDFR	12,185 / 606	10 / 3	$\sim$3.0 / $\sim$0.3	$\sim$24 GB / $\sim$32 GB
Weather	STIT / RDFR	36,792 / 1,830	4 / 2	$\sim$3.7 / $\sim$0.5	$\sim$24 GB / $\sim$32 GB

Table 7: Dataset statistics are from [Wu et al., 2023]. The dimension indicates the number of time series (i.e., channels), and the dataset size is organized in (training, validation, testing).

Tasks	Dataset	Dim.	Series Length	Dataset Size	Frequency	Domain
Long-term Forecasting	ETTm1	7	{96, 192, 336, 720}	(34465, 11521, 11521)	15 min	Temperature
	ETTm2	7	{96, 192, 336, 720}	(34465, 11521, 11521)	15 min	Temperature
	ETTh1	7	{96, 192, 336, 720}	(8545, 2881, 2881)	1 hour	Temperature
	ETTh2	7	{96, 192, 336, 720}	(8545, 2881, 2881)	1 hour	Temperature
	Electricity	321	{96, 192, 336, 720}	(18317, 2633, 5261)	1 hour	Electricity
	Traffic	862	{96, 192, 336, 720}	(12185, 1757, 3509)	1 hour	Transportation
	Weather	21	{96, 192, 336, 720}	(36792, 5271, 10540)	10 min	Weather
	ILI	7	{24, 36, 48, 60}	(617, 74, 170)	1 week	Illness
Short-term Forecasting	M4-Yearly	1	6	(23000, 0, 23000)	Yearly	Demographic
	M4-Quarterly	1	8	(24000, 0, 24000)	Quarterly	Finance
	M4-Monthly	1	18	(48000, 0, 48000)	Monthly	Industry
	M4-Weekly	1	13	(359, 0, 359)	Weekly	Macro
	M4-Daily	1	14	(4227, 0, 4227)	Daily	Micro
	M4-Hourly	1	48	(414, 0, 414)	Hourly	Other

Table 8: Statistical significance analysis on representative long-term forecasting settings with $H = 96$. We report mean and standard deviation over three independent runs with different random seeds. p-values are computed using a paired two-sided t-test between RDTU and Time-LLM over matched dataset-seed pairs.

Dataset	**Metric**	**RDTU**	**Time-LLM**	**Relative Gain**	**p-value**
ETTh1	MSE / MAE	0.352 ± 0.003 / 0.383 ± 0.002	0.376 ± 0.004 / 0.403 ± 0.003	6.38% / 4.96%	0.018 / 0.021
Electricity	MSE / MAE	0.125 ± 0.002 / 0.211 ± 0.002	0.137 ± 0.002 / 0.233 ± 0.003	8.76% / 9.44%	0.014 / 0.012
Traffic	MSE / MAE	0.359 ± 0.004 / 0.236 ± 0.003	0.393 ± 0.005 / 0.268 ± 0.004	8.65% / 11.94%	0.011 / 0.009
Average	MSE / MAE	0.279 ± 0.003 / 0.277 ± 0.002	0.302 ± 0.004 / 0.301 ± 0.003	7.62% / 7.97%	0.013 / 0.015

Table 9: Statistical significance analysis of the RDFR stage on ETTh1 with $H = 96$. We report mean and standard deviation over three independent runs. p-values are computed using a paired two-sided t-test between STIT and RDTU.

Method	**MSE** $\downarrow$	**MAE** $\downarrow$	**MSE Gain**	**MAE Gain**
Qwen2.5-Instruct-STIT	0.403 ± 0.002	0.428 ± 0.002	–	–
RDTU w/ RDFR	$\mathbf{0.351} \pm \mathbf{0.002}$	$\mathbf{0.382} \pm \mathbf{0.002}$	12.90%	10.75%
p-value	0.004	0.006	–	–

Table 10: Full long-term forecasting results on the benchmark datasets. We followed the same setting in DMMV [Shen et al., 2025]. Lower MSE and MAE indicate better performance. We set the forecasting horizons $H \in \{24, 36, 48, 60\}$ for Illness and $\{96, 192, 336, 720\}$ for the others. **Red**: the best, Blue: the second best.

View		**Language**						**Multi-Modal**				**Visual**		**Numerical**									
Model		RDTU		GPT4TS		Time-LLM		DMMV-A		Time-VLM		VisionTS		PatchTST		CycleNet		TimesNet		DLinear		FEDformer	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	**0.351**	**0.382**	0.370	0.389	0.376	0.402	0.354	0.389	0.361	0.386	0.355	0.386	0.370	0.399	0.374	0.396	0.384	0.402	0.375	0.399	0.376	0.419
	192	**0.389**	**0.403**	0.412	0.413	0.407	0.421	0.393	0.405	0.397	0.415	0.395	0.407	0.413	0.421	0.406	0.415	0.436	0.429	0.405	0.416	0.420	0.448
	336	0.412	0.419	0.448	0.431	0.430	0.438	**0.387**	**0.413**	0.420	0.421	0.419	0.421	0.422	0.436	0.431	0.430	0.491	0.469	0.439	0.416	0.459	0.465
	720	0.442	**0.446**	**0.441**	0.449	0.457	0.468	0.445	0.450	**0.441**	0.458	0.458	0.460	0.447	0.466	0.450	0.464	0.521	0.500	0.472	0.490	0.506	0.507
	Avg.	0.399	**0.413**	0.418	0.421	0.418	0.432	**0.395**	0.414	0.405	0.420	0.407	0.419	0.413	0.431	0.415	0.426	0.458	0.450	0.423	0.430	0.440	0.460
ETTh2	96	**0.263**	**0.329**	0.280	0.335	0.286	0.346	0.294	0.349	0.267	0.335	0.288	0.334	0.274	0.336	0.279	0.341	0.340	0.374	0.289	0.353	0.358	0.397
	192	0.334	**0.372**	0.348	0.380	0.361	0.391	0.339	0.395	**0.326**	0.373	0.349	0.380	0.339	0.379	0.342	0.385	0.402	0.414	0.383	0.418	0.429	0.439
	336	0.355	0.381	0.380	0.405	0.390	0.414	**0.322**	0.384	0.357	0.406	0.364	0.398	0.329	**0.380**	0.371	0.413	0.452	0.452	0.448	0.465	0.496	0.487
	720	0.395	0.425	0.406	0.436	0.405	0.434	0.392	0.425	0.412	0.449	0.403	0.431	**0.379**	**0.422**	0.426	0.451	0.462	0.468	0.605	0.551	0.463	0.474
	Avg.	0.337	**0.377**	0.354	0.389	0.361	0.396	0.337	0.388	0.341	0.391	0.351	0.386	**0.330**	0.379	0.355	0.398	0.414	0.427	0.431	0.447	0.437	0.449
ETTm1	96	**0.272**	**0.326**	0.300	0.340	0.291	0.341	0.279	0.329	0.304	0.346	0.284	0.332	0.290	0.342	0.299	0.348	0.338	0.375	0.299	0.343	0.379	0.419
	192	0.327	**0.354**	0.343	0.368	0.341	0.369	**0.317**	0.357	0.332	0.366	0.327	0.362	0.332	0.369	0.334	0.367	0.374	0.387	0.335	0.365	0.426	0.441
	336	**0.350**	**0.375**	0.376	0.386	0.359	0.379	0.351	0.381	0.364	0.383	0.354	0.382	0.366	0.392	0.368	0.386	0.410	0.411	0.369	0.386	0.445	0.459
	720	0.410	0.412	0.431	0.416	0.433	0.419	0.411	0.415	**0.402**	**0.410**	0.411	0.415	0.416	0.420	0.417	0.414	0.478	0.450	0.425	0.421	0.543	0.490
	Avg.	**0.340**	**0.367**	0.363	0.378	0.356	0.377	**0.340**	0.371	0.351	0.376	0.344	0.373	0.351	0.381	0.355	0.379	0.400	0.406	0.357	0.379	0.448	0.452
ETTm2	96	**0.157**	**0.244**	0.163	0.249	0.162	0.248	0.172	0.260	0.160	0.250	0.174	0.262	0.165	0.255	0.159	0.247	0.187	0.267	0.167	0.260	0.203	0.287
	192	0.219	0.288	0.222	0.291	0.235	0.304	0.227	0.298	0.215	0.291	0.228	0.297	0.220	0.292	**0.214**	**0.286**	0.249	0.309	0.224	0.303	0.269	0.328
	336	0.270	0.323	0.273	0.327	0.280	0.329	0.272	0.327	0.270	0.325	0.281	0.337	0.274	0.329	**0.269**	**0.322**	0.321	0.351	0.281	0.342	0.325	0.366
	720	0.354	0.379	0.357	**0.376**	0.366	0.382	0.351	0.381	**0.348**	0.378	0.384	0.410	0.362	0.385	0.363	0.382	0.408	0.403	0.397	0.421	0.421	0.415
	Avg.	0.250	**0.309**	0.254	0.311	0.261	0.316	0.256	0.317	**0.248**	0.311	0.267	0.327	0.255	0.315	0.251	**0.309**	0.291	0.333	0.267	0.332	0.305	0.349
Illness	24	1.609	0.771	1.869	0.823	1.792	0.807	1.409	**0.754**	–	–	1.613	0.834	**1.319**	**0.754**	2.255	1.017	2.317	0.934	2.215	1.081	3.228	1.260
	36	1.453	0.835	1.853	0.854	1.833	0.833	**1.290**	**0.745**	–	–	1.316	0.750	1.430	0.834	2.121	0.950	1.972	0.920	1.963	0.963	2.679	1.080
	48	1.545	0.841	1.886	0.855	2.269	1.012	**1.499**	**0.810**	–	–	1.548	0.818	1.553	0.815	2.187	1.007	2.238	0.940	2.130	1.024	2.622	1.078
	60	1.630	0.802	1.877	0.877	2.177	0.925	**1.428**	**0.773**	–	–	1.450	0.783	1.470	0.788	2.185	0.997	2.027	0.928	2.368	1.096	2.857	1.157
	Avg.	1.559	0.812	1.871	0.852	2.018	0.894	**1.407**	**0.771**	–	–	1.482	0.796	1.443	0.798	2.187	0.992	2.139	0.931	2.169	1.041	2.847	1.144
Electricity	96	**0.124**	**0.210**	0.141	0.239	0.137	0.233	0.126	0.213	0.142	0.245	0.127	0.217	0.129	0.222	0.128	0.223	0.168	0.272	0.140	0.237	0.193	0.308
	192	**0.144**	**0.235**	0.158	0.253	0.152	0.247	0.145	0.237	0.157	0.260	0.148	0.237	0.157	0.240	**0.144**	0.237	0.184	0.289	0.153	0.249	0.201	0.315
	336	0.162	0.257	0.172	0.266	0.169	0.267	0.162	0.254	0.174	0.276	0.163	**0.253**	0.163	0.259	0.160	0.254	0.198	0.300	0.169	0.267	0.214	0.329
	720	**0.193**	**0.281**	0.207	0.293	0.200	0.290	0.197	0.286	0.214	0.308	0.199	0.293	0.197	0.290	0.198	0.287	0.220	0.320	0.203	0.301	0.246	0.355
	Avg.	**0.156**	**0.246**	0.170	0.263	0.165	0.259	0.158	0.248	0.172	0.272	0.159	0.250	0.162	0.253	0.158	0.250	0.193	0.295	0.166	0.264	0.214	0.327
Weather	96	**0.141**	**0.184**	0.148	0.188	0.155	0.199	0.143	0.195	0.148	0.200	0.146	0.191	0.149	0.198	0.167	0.221	0.172	0.220	0.176	0.237	0.217	0.296
	192	0.189	**0.227**	0.192	0.230	0.223	0.261	**0.187**	0.242	0.193	0.240	0.194	0.238	0.194	0.241	0.212	0.258	0.219	0.261	0.220	0.282	0.276	0.336
	336	0.239	**0.272**	0.246	0.273	0.251	0.279	**0.237**	0.273	0.243	0.281	0.243	0.275	0.245	0.282	0.260	0.293	0.280	0.306	0.265	0.319	0.339	0.380
	720	0.314	0.322	0.320	0.328	0.345	0.342	**0.302**	**0.315**	0.312	0.332	0.318	0.328	0.314	0.334	0.328	0.339	0.365	0.359	0.333	0.362	0.403	0.428
	Avg.	0.221	**0.251**	0.227	0.255	0.244	0.270	**0.217**	0.256	0.224	0.263	0.225	0.258	0.226	0.264	0.242	0.278	0.259	0.287	0.249	0.300	0.309	0.360
Traffic	96	0.358	0.235	0.396	0.264	0.392	0.267	**0.344**	0.237	0.393	0.290	0.346	**0.232**	0.360	0.249	0.397	0.278	0.593	0.321	0.410	0.282	0.587	0.366
	192	0.375	0.248	0.412	0.268	0.409	0.271	**0.363**	0.249	0.405	0.296	0.376	**0.245**	0.379	0.256	0.411	0.283	0.617	0.336	0.423	0.287	0.604	0.373
	336	0.388	0.256	0.421	0.273	0.434	0.296	**0.387**	0.256	0.420	0.305	0.389	**0.252**	0.392	0.264	0.424	0.289	0.629	0.336	0.436	0.296	0.621	0.383
	720	0.435	**0.280**	0.455	0.291	0.451	0.291	0.433	0.284	0.459	0.323	**0.432**	0.293	**0.432**	0.286	0.450	0.305	0.640	0.350	0.466	0.315	0.626	0.382
	Avg.	0.389	**0.255**	0.421	0.274	0.422	0.281	0.389	0.257	0.419	0.304	**0.386**	0.256	0.391	0.264	0.421	0.289	0.620	0.336	0.434	0.295	0.610	0.376
# Wins		**35**		2		0		23		6		6		7		6		0		0		0	

Table 11: Additional long-term forecasting results. We set the forecasting horizons $H \in \{24, 36, 48, 60\}$ for ILI and $\{96, 192, 336, 720\}$ for the others. A lower value indicates better performance. **Red**: the best, Blue: the second best.

Methods		RDTU		Autoformer		Stationary		ETSformer		LightTS		Informer		Reformer	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	**0.351**	**0.382**	0.449	0.459	0.513	0.491	0.494	0.479	0.424	0.432	0.865	0.713	0.837	0.728
	192	**0.389**	**0.403**	0.500	0.482	0.534	0.504	0.538	0.504	0.475	0.462	1.008	0.792	0.923	0.766
	336	**0.412**	**0.419**	0.521	0.496	0.588	0.535	0.574	0.521	0.518	0.488	1.107	0.809	1.097	0.835
	720	**0.442**	**0.446**	0.514	0.512	0.643	0.616	0.562	0.535	0.547	0.533	1.181	0.865	1.257	0.889
	Avg	**0.399**	**0.413**	0.496	0.487	0.570	0.537	0.542	0.510	0.491	0.479	1.040	0.795	1.029	0.805
ETTh2	96	**0.263**	**0.329**	0.346	0.388	0.476	0.458	0.340	0.391	0.397	0.437	3.755	1.525	2.626	1.317
	192	**0.334**	**0.372**	0.456	0.452	0.512	0.493	0.430	0.439	0.520	0.504	5.602	1.931	11.12	2.979
	336	**0.355**	**0.381**	0.482	0.486	0.552	0.551	0.485	0.479	0.626	0.559	4.721	1.835	9.323	2.769
	720	**0.395**	**0.425**	0.515	0.511	0.562	0.560	0.500	0.497	0.863	0.672	3.647	1.625	3.874	1.697
	Avg	**0.337**	**0.377**	0.450	0.459	0.526	0.516	0.439	0.452	0.602	0.543	4.431	1.729	6.736	2.191
ETTm1	96	**0.272**	**0.326**	0.505	0.475	0.386	0.398	0.375	0.398	0.374	0.400	0.672	0.571	0.538	0.528
	192	**0.327**	**0.354**	0.553	0.496	0.459	0.444	0.408	0.410	0.400	0.407	0.795	0.669	0.658	0.592
	336	**0.350**	**0.375**	0.621	0.537	0.495	0.464	0.435	0.428	0.438	0.438	1.212	0.871	0.898	0.721
	720	**0.410**	**0.412**	0.671	0.561	0.585	0.516	0.499	0.462	0.527	0.502	1.166	0.823	1.102	0.841
	Avg	**0.340**	**0.367**	0.588	0.517	0.481	0.456	0.429	0.425	0.435	0.437	0.961	0.734	0.799	0.671
ETTm2	96	**0.157**	**0.244**	0.255	0.339	0.192	0.274	0.189	0.280	0.209	0.308	0.365	0.453	0.658	0.619
	192	**0.219**	**0.288**	0.281	0.340	0.280	0.339	0.253	0.319	0.311	0.382	0.533	0.563	1.078	0.827
	336	**0.270**	**0.323**	0.339	0.372	0.334	0.361	0.314	0.357	0.442	0.466	1.363	0.887	1.549	0.972
	720	**0.354**	**0.379**	0.433	0.432	0.417	0.413	0.414	0.413	0.675	0.587	3.379	1.338	2.631	1.242
	Avg	**0.250**	**0.309**	0.327	0.371	0.306	0.347	0.293	0.342	0.409	0.436	1.410	0.810	1.479	0.915
Weather	96	**0.141**	**0.184**	0.266	0.336	0.173	0.223	0.197	0.281	0.182	0.242	0.300	0.384	0.689	0.596
	192	**0.189**	**0.227**	0.307	0.367	0.245	0.285	0.237	0.312	0.227	0.287	0.598	0.544	0.752	0.638
	336	**0.239**	**0.272**	0.359	0.395	0.321	0.338	0.298	0.353	0.282	0.334	0.578	0.523	0.639	0.596
	720	**0.314**	**0.322**	0.419	0.428	0.414	0.410	0.352	0.288	0.352	0.386	1.059	0.741	1.130	0.792
	Avg	**0.221**	**0.251**	0.338	0.382	0.288	0.314	0.271	0.334	0.261	0.312	0.634	0.548	0.803	0.656
Electricity	96	**0.124**	**0.210**	0.201	0.317	0.169	0.273	0.187	0.304	0.207	0.307	0.274	0.368	0.312	0.402
	192	**0.144**	**0.235**	0.222	0.334	0.182	0.286	0.199	0.315	0.213	0.316	0.296	0.386	0.348	0.433
	336	**0.162**	**0.257**	0.231	0.338	0.200	0.304	0.212	0.329	0.230	0.333	0.300	0.394	0.350	0.433
	720	**0.193**	**0.281**	0.254	0.361	0.222	0.321	0.233	0.345	0.265	0.360	0.373	0.439	0.340	0.420
	Avg	**0.156**	**0.246**	0.227	0.338	0.193	0.296	0.208	0.323	0.229	0.329	0.311	0.397	0.338	0.422
Traffic	96	**0.358**	**0.235**	0.613	0.388	0.612	0.338	0.607	0.392	0.615	0.391	0.719	0.391	0.732	0.423
	192	**0.375**	**0.248**	0.616	0.382	0.613	0.340	0.621	0.399	0.601	0.382	0.696	0.379	0.733	0.420
	336	**0.388**	**0.256**	0.622	0.337	0.618	0.328	0.622	0.396	0.613	0.386	0.777	0.420	0.742	0.420
	720	**0.435**	**0.280**	0.660	0.408	0.653	0.355	0.632	0.396	0.658	0.407	0.864	0.472	0.755	0.423
	Avg	**0.389**	**0.255**	0.628	0.379	0.624	0.340	0.621	0.396	0.622	0.392	0.764	0.416	0.741	0.422
ILI	24	**1.609**	**0.771**	3.483	1.287	2.294	0.945	2.527	1.020	8.313	2.144	5.764	1.677	4.400	1.382
	36	**1.453**	**0.835**	3.103	1.148	1.825	0.848	2.615	1.007	6.631	1.902	4.755	1.467	4.783	1.448
	48	**1.545**	**0.841**	2.669	1.085	2.010	0.900	2.359	0.972	7.299	1.982	4.763	1.469	4.832	1.465
	60	**1.630**	**0.802**	2.770	1.125	2.178	0.963	2.487	1.016	7.283	1.985	5.264	1.564	4.882	1.483
	Avg	**1.559**	**0.812**	3.006	1.161	2.077	0.914	2.497	1.004	7.382	2.003	5.137	1.544	4.724	1.445
1^{st} Count		40		0		0		0		0		0		0	

Table 12: Full short-term time series forecasting results. The forecasting horizons are in [6, 48] and the last three rows are weighted averaged from all datasets under different sampling intervals. A lower value indicates better performance. **Red**: the best, Blue: the second best.

	Methods	RDTU	Time-LLM	GPT4TS	TimesNet	PatchTST	N-HiTS	N-BEATS	ETSformer	LightTS	DLinear	FEDformer	Stationary	Autoformer	Informer	Reformer
Yearly	SMAPE	**13.408**	13.419	15.11	15.378	13.477	13.422	13.487	18.009	14.247	16.965	14.021	13.717	13.974	14.727	16.169
	MASE	3.014	**3.005**	3.565	3.554	3.019	3.056	3.036	4.487	3.109	4.283	3.036	3.078	3.134	3.418	3.800
	OWA	**0.788**	0.789	0.911	0.918	0.792	0.795	0.795	1.115	0.827	1.058	0.811	0.807	0.822	0.881	0.973
Quarterly	SMAPE	**10.094**	10.110	10.597	10.465	10.38	10.185	10.564	13.376	11.364	12.145	11.1	10.958	11.338	11.360	13.313
	MASE	**1.176**	1.178	1.253	1.227	1.233	1.18	1.252	1.906	1.328	1.520	1.35	1.325	1.365	1.401	1.775
	OWA	**0.888**	0.889	0.938	0.923	0.921	0.893	0.936	1.302	1.000	1.106	0.996	0.981	1.012	1.027	1.252
Monthly	SMAPE	12.977	12.980	13.258	13.513	**12.959**	13.059	13.089	14.588	14.014	13.514	14.403	13.917	13.958	14.062	20.128
	MASE	0.968	**0.963**	1.003	1.039	0.97	1.013	0.996	1.368	1.053	1.037	1.147	1.097	1.103	1.141	2.614
	OWA	**0.903**	**0.903**	0.931	0.957	0.905	0.929	0.922	1.149	0.981	0.956	1.038	0.998	1.002	1.024	1.927
Others	SMAPE	4.730	4.795	6.124	6.913	4.952	**4.711**	6.599	7.267	15.880	6.709	7.148	6.302	5.485	24.460	32.491
	MASE	3.086	3.178	4.116	4.507	3.347	**3.054**	4.43	5.240	11.434	4.953	4.041	4.064	3.865	20.960	33.355
	OWA	0.985	1.006	1.259	1.438	1.049	**0.977**	1.393	1.591	3.474	1.487	1.389	1.304	1.187	5.879	8.679
Average	SMAPE	**11.979**	11.983	12.69	12.88	12.059	12.035	12.25	14.718	13.525	13.639	13.16	12.780	12.909	14.086	18.200
	MASE	1.599	**1.595**	1.808	1.836	1.623	1.625	1.698	2.408	2.111	2.095	1.775	1.756	1.771	2.718	4.223
	OWA	**0.859**	**0.859**	0.94	0.955	0.869	0.869	0.896	1.172	1.051	1.051	0.949	0.930	0.939	1.230	1.775

Table 13: Full few-shot learning results on 10% training data.

Methods		RDTU		Time-LLM		GPT4TS		DLinear		PatchTST		TimesNet		FEDformer		Autoformer		Stationary		ETSformer		LightTS		Informer		Reformer	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	**0.443**	**0.451**	0.448	0.460	0.458	0.456	0.492	0.495	0.516	0.485	0.861	0.628	0.512	0.499	0.613	0.552	0.918	0.639	1.112	0.806	1.298	0.838	1.179	0.792	1.184	0.790
	192	**0.482**	**0.479**	0.484	0.483	0.570	0.516	0.565	0.538	0.598	0.524	0.797	0.593	0.624	0.555	0.722	0.598	0.915	0.629	1.155	0.823	1.322	0.854	1.199	0.806	1.295	0.850
	336	**0.585**	0.536	0.589	0.540	0.608	**0.535**	0.721	0.622	0.657	0.550	0.941	0.648	0.691	0.574	0.750	0.619	0.939	0.644	1.179	0.832	1.347	0.870	1.202	0.811	1.294	0.854
	720	**0.694**	**0.587**	0.700	0.604	0.725	0.591	0.986	0.743	0.762	0.610	0.877	0.641	0.728	0.614	0.721	0.616	0.887	0.645	1.273	0.874	1.534	0.947	1.217	0.825	1.223	0.838
	Avg	**0.551**	**0.513**	0.556	0.522	0.590	0.525	0.691	0.600	0.633	0.542	0.869	0.628	0.639	0.561	0.702	0.596	0.915	0.639	1.180	0.834	1.375	0.877	1.199	0.809	1.249	0.833
ETTh2	96	**0.271**	**0.322**	0.275	0.326	0.331	0.374	0.357	0.411	0.353	0.389	0.378	0.409	0.382	0.416	0.413	0.451	0.389	0.411	0.678	0.619	2.022	1.006	3.837	1.508	3.788	1.533
	192	**0.369**	**0.367**	0.374	0.373	0.402	0.411	0.569	0.519	0.403	0.414	0.490	0.467	0.478	0.474	0.474	0.477	0.473	0.455	0.785	0.666	2.329	1.104	3.856	1.513	3.552	1.483
	336	**0.402**	**0.423**	0.406	0.429	0.406	0.433	0.671	0.572	0.426	0.441	0.537	0.494	0.504	0.501	0.547	0.543	0.507	0.480	0.839	0.694	2.453	1.122	3.952	1.526	3.395	1.526
	720	**0.425**	**0.443**	0.427	0.449	0.449	0.464	0.824	0.648	0.477	0.480	0.510	0.491	0.499	0.509	0.516	0.523	0.477	0.472	1.273	0.874	3.816	1.407	3.842	1.503	3.205	1.401
	Avg	**0.367**	**0.389**	0.370	0.394	0.397	0.421	0.605	0.538	0.415	0.431	0.479	0.465	0.466	0.475	0.488	0.499	0.462	0.455	0.894	0.713	2.655	1.160	3.872	1.513	3.485	1.486
ETTm1	96	**0.343**	**0.386**	0.346	0.388	0.390	0.404	0.352	0.392	0.410	0.419	0.583	0.501	0.578	0.518	0.774	0.614	0.761	0.568	0.911	0.688	0.921	0.682	1.162	0.785	1.442	0.847
	192	0.375	**0.411**	**0.373**	0.416	0.429	0.423	0.382	0.412	0.437	0.434	0.630	0.528	0.617	0.546	0.754	0.592	0.781	0.574	0.955	0.703	0.957	0.701	1.172	0.793	1.444	0.862
	336	**0.410**	**0.424**	0.413	0.426	0.469	0.439	0.419	0.434	0.476	0.454	0.725	0.568	0.998	0.775	0.869	0.677	0.803	0.587	0.991	0.719	0.998	0.716	1.227	0.908	1.450	0.866
	720	**0.478**	**0.473**	0.485	0.476	0.569	0.498	0.490	0.477	0.681	0.556	0.769	0.549	0.693	0.579	0.810	0.630	0.844	0.581	1.062	0.747	1.007	0.719	1.207	0.797	1.366	0.850
	Avg	**0.402**	**0.424**	0.404	0.427	0.464	0.441	0.411	0.429	0.501	0.466	0.677	0.537	0.722	0.605	0.802	0.628	0.797	0.578	0.980	0.714	0.971	0.705	1.192	0.821	1.426	0.856
ETTm2	96	**0.174**	**0.260**	0.177	0.261	0.188	0.269	0.213	0.303	0.191	0.274	0.212	0.285	0.291	0.399	0.352	0.454	0.229	0.308	0.331	0.430	0.813	0.688	3.203	1.407	4.195	1.628
	192	**0.236**	**0.308**	0.241	0.314	0.251	0.309	0.278	0.345	0.252	0.317	0.270	0.323	0.307	0.379	0.694	0.691	0.291	0.343	0.400	0.464	1.008	0.768	3.112	1.387	4.042	1.601
	336	**0.272**	**0.323**	0.274	0.327	0.307	0.346	0.338	0.385	0.306	0.353	0.323	0.353	0.543	0.559	2.408	1.407	0.348	0.376	0.469	0.498	1.031	0.775	3.255	1.421	3.963	1.585
	720	**0.413**	**0.384**	0.417	0.390	0.426	0.417	0.436	0.440	0.433	0.427	0.474	0.449	0.712	0.614	1.913	1.166	0.461	0.438	0.589	0.557	1.096	0.791	3.909	1.543	3.711	1.532
	Avg	**0.274**	**0.319**	0.277	0.323	0.293	0.335	0.316	0.368	0.296	0.343	0.320	0.353	0.463	0.488	1.342	0.930	0.332	0.366	0.447	0.487	0.987	0.756	3.370	1.440	3.978	1.587
Weather	96	**0.159**	**0.209**	0.161	0.210	0.163	0.215	0.171	0.224	0.165	0.215	0.184	0.230	0.188	0.253	0.221	0.297	0.192	0.234	0.199	0.272	0.217	0.269	0.374	0.401	0.335	0.380
	192	**0.199**	**0.244**	0.204	0.248	0.210	0.254	0.215	0.263	0.210	0.257	0.245	0.283	0.250	0.304	0.270	0.322	0.269	0.295	0.279	0.332	0.259	0.304	0.552	0.478	0.522	0.462
	336	0.257	**0.292**	0.261	0.302	**0.256**	**0.292**	0.258	0.299	0.259	0.297	0.305	0.321	0.312	0.346	0.320	0.351	0.370	0.357	0.356	0.386	0.303	0.334	724	0.541	0.715	0.535
	720	0.311	0.335	**0.309**	**0.332**	0.321	0.339	0.320	0.346	0.332	0.346	0.381	0.371	0.387	0.393	0.390	0.396	0.441	0.405	0.437	0.448	0.377	0.382	0.739	0.558	0.611	0.500
	Avg	**0.232**	**0.270**	0.234	0.273	0.238	0.275	0.241	0.283	0.242	0.279	0.279	0.301	0.284	0.324	0.300	0.342	0.318	0.323	0.318	0.360	0.289	0.322	0.597	0.495	0.546	0.469
Electricity	96	0.141	**0.237**	**0.139**	0.241	**0.139**	**0.237**	0.150	0.253	0.140	0.238	0.299	0.373	0.231	0.323	0.261	0.348	0.420	0.466	0.599	0.587	0.350	0.425	1.259	0.919	0.993	0.784
	192	0.153	**0.247**	**0.151**	0.248	0.156	0.252	0.164	0.264	0.160	0.255	0.305	0.379	0.261	0.356	0.338	0.406	0.411	0.459	0.620	0.598	0.376	0.448	1.160	0.873	0.938	0.753
	336	0.171	**0.269**	**0.169**	0.270	0.175	0.270	0.181	0.282	0.180	0.276	0.319	0.391	0.360	0.445	0.410	0.474	0.434	0.473	0.662	0.619	0.428	0.485	1.157	0.872	0.925	0.745
	720	0.227	0.318	0.240	0.322	0.233	**0.317**	**0.223**	0.321	0.241	0.323	0.369	0.426	0.530	0.585	0.715	0.685	0.510	0.521	0.757	0.664	0.611	0.597	1.203	0.898	1.004	0.790
	Avg	**0.173**	**0.268**	0.175	0.270	0.176	0.269	0.180	0.280	0.180	0.273	0.323	0.392	0.346	0.427	0.431	0.478	0.444	0.480	0.660	0.617	0.441	0.489	1.195	0.891	0.965	0.768
Traffic	96	0.405	**0.288**	0.418	0.291	0.414	0.297	0.419	0.298	**0.403**	0.289	0.719	0.416	0.639	0.400	0.672	0.405	1.412	0.802	1.643	0.855	1.157	0.636	1.557	0.821	1.527	0.815
	192	**0.412**	**0.292**	0.414	0.296	0.426	0.301	0.434	0.305	0.415	0.296	0.748	0.428	0.637	0.416	0.727	0.424	1.419	0.806	1.641	0.854	1.207	0.661	1.454	0.765	1.538	0.817
	336	0.424	0.306	**0.421**	0.311	0.434	**0.303**	0.449	0.313	0.426	0.304	0.853	0.471	0.655	0.427	0.749	0.454	1.443	0.815	1.711	0.878	1.334	0.713	1.521	0.812	1.550	0.819
	720	0.463	**0.325**	**0.462**	0.327	0.487	0.337	0.484	0.336	0.474	0.331	1.485	0.825	0.722	0.456	0.847	0.499	1.539	0.837	2.660	1.157	1.292	0.726	1.605	0.846	1.588	0.833
	Avg	**0.426**	**0.303**	0.429	0.306	0.440	0.310	0.447	0.313	0.430	0.305	0.951	0.535	0.663	0.425	0.749	0.446	1.453	0.815	1.914	0.936	1.248	0.684	1.534	0.811	1.551	0.821
1^{st} Count		**58**		8		7		1		1		0		0		0		0		0		0		0		0	

Table 14: Full few-shot learning results on 5% training data. '-' means that 5% time series is not sufficient to constitute a training set.

Methods		RDTU		Time-LLM		GPT4TS		DLinear		PatchTST		TimesNet		FEDformer		Autoformer		Stationary		ETSformer		LightTS		Informer		Reformer	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	**0.479**	**0.460**	0.483	0.464	0.543	0.506	0.547	0.503	0.557	0.519	0.892	0.625	0.593	0.529	0.681	0.570	0.952	0.650	1.169	0.832	1.483	0.91	1.225	0.812	1.198	0.795
	192	**0.624**	**0.536**	0.629	0.540	0.748	0.580	0.720	0.604	0.711	0.570	0.940	0.665	0.652	0.563	0.725	0.602	0.943	0.645	1.221	0.853	1.525	0.93	1.249	0.828	1.273	0.853
	336	0.741	0.603	0.768	0.626	0.754	0.595	0.984	0.727	0.816	0.619	0.945	0.653	**0.731**	**0.594**	0.761	0.624	0.935	0.644	1.179	0.832	1.347	0.87	1.202	0.811	1.254	0.857
	720	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Avg	**0.615**	**0.535**	0.627	0.543	0.681	0.560	0.750	0.611	0.694	0.569	0.925	0.647	0.658	0.562	0.722	0.598	0.943	0.646	1.189	0.839	1.451	0.903	1.225	0.817	1.241	0.835
ETTh2	96	**0.332**	**0.394**	0.336	0.397	0.376	0.421	0.442	0.456	0.401	0.421	0.409	0.420	0.390	0.424	0.428	0.468	0.408	0.423	0.678	0.619	2.022	1.006	3.837	1.508	3.753	1.518
	192	**0.405**	**0.423**	0.406	0.425	0.418	0.441	0.617	0.542	0.452	0.455	0.483	0.464	0.457	0.465	0.496	0.504	0.497	0.468	0.845	0.697	3.534	1.348	3.975	1.933	3.516	1.473
	336	0.410	0.438	**0.405**	**0.432**	0.408	0.439	1.424	0.849	0.464	0.469	0.499	0.479	0.477	0.483	0.486	0.496	0.507	0.481	0.905	0.727	4.063	1.451	3.956	1.520	3.312	1.427
	720	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Avg	**0.382**	**0.418**	**0.382**	**0.418**	0.400	0.433	0.694	0.577	0.827	0.615	0.439	0.448	0.463	0.454	0.441	0.457	0.470	0.489	0.809	0.681	3.206	1.268	3.922	1.653	3.527	1.472
ETTm1	96	**0.313**	**0.369**	0.316	0.377	0.386	0.405	0.332	0.374	0.399	0.414	0.606	0.518	0.628	0.544	0.726	0.578	0.823	0.587	1.031	0.747	1.048	0.733	1.130	0.775	1.234	0.798
	192	0.438	0.427	0.450	0.464	0.440	0.438	**0.358**	**0.390**	0.441	0.436	0.681	0.539	0.666	0.566	0.750	0.591	0.844	0.591	1.087	0.766	1.097	0.756	1.150	0.788	1.287	0.839
	336	0.446	0.431	0.450	0.424	0.485	0.459	**0.402**	**0.416**	0.499	0.467	0.786	0.597	0.807	0.628	0.851	0.659	0.870	0.603	1.138	0.787	1.147	0.775	1.198	0.809	1.288	0.842
	720	0.485	0.478	**0.483**	**0.471**	0.577	0.499	0.511	0.489	0.767	0.587	0.796	0.593	0.822	0.633	0.857	0.655	0.893	0.611	1.245	0.831	1.200	0.799	1.175	0.794	1.247	0.828
	Avg	0.421	0.426	0.425	0.434	0.472	0.450	**0.400**	**0.417**	0.526	0.476	0.717	0.561	0.730	0.592	0.796	0.620	0.857	0.598	1.125	0.782	1.123	0.765	1.163	0.791	1.264	0.826
ETTm2	96	0.176	**0.260**	**0.174**	0.261	0.199	0.280	0.236	0.326	0.206	0.288	0.220	0.299	0.229	0.320	0.232	0.322	0.238	0.316	0.404	0.485	1.108	0.772	3.599	1.478	3.883	1.545
	192	**0.213**	**0.286**	0.215	0.287	0.256	0.316	0.306	0.373	0.264	0.324	0.311	0.361	0.394	0.361	0.291	0.357	0.298	0.349	0.479	0.521	1.317	0.850	3.578	1.475	3.553	1.484
	336	**0.271**	**0.327**	0.273	0.330	0.318	0.353	0.380	0.423	0.334	0.367	0.338	0.366	0.378	0.427	0.478	0.517	0.353	0.380	0.552	0.555	1.415	0.879	3.561	1.473	3.446	1.460
	720	0.436	0.415	**0.433**	**0.412**	0.460	0.436	0.674	0.583	0.454	0.432	0.509	0.465	0.523	0.510	0.553	0.538	0.475	0.445	0.701	0.627	1.822	0.984	3.896	1.533	3.445	1.460
	Avg	**0.274**	**0.322**	**0.274**	0.323	0.308	0.346	0.399	0.426	0.314	0.352	0.344	0.372	0.381	0.404	0.388	0.433	0.341	0.372	0.534	0.547	1.415	0.871	3.658	1.489	3.581	1.487
Weather	96	**0.168**	**0.220**	0.172	0.263	0.175	0.230	0.184	0.242	0.171	0.224	0.207	0.253	0.229	0.309	0.227	0.299	0.215	0.252	0.218	0.295	0.230	0.285	0.497	0.497	0.406	0.435
	192	**0.220**	**0.267**	0.224	0.271	0.227	0.276	0.228	0.283	0.230	0.277	0.272	0.307	0.265	0.317	0.278	0.333	0.290	0.307	0.294	0.331	0.274	0.323	0.620	0.545	0.446	0.450
	336	**0.278**	**0.317**	0.282	0.321	0.286	0.322	0.279	0.322	0.294	0.326	0.313	0.328	0.353	0.392	0.351	0.393	0.353	0.348	0.359	0.398	0.318	0.355	0.649	0.547	0.465	0.459
	720	0.367	0.384	0.366	0.381	0.366	**0.379**	**0.364**	0.388	0.384	0.387	0.400	0.385	0.391	0.394	0.387	0.389	0.452	0.407	0.461	0.461	0.401	0.418	0.570	0.522	0.471	0.468
	Avg	**0.258**	**0.297**	0.260	0.309	0.263	0.301	0.263	0.308	0.269	0.303	0.298	0.318	0.309	0.353	0.310	0.353	0.327	0.328	0.333	0.371	0.305	0.345	0.584	0.527	0.447	0.453
Electricity	96	**0.143**	**0.238**	0.147	0.242	**0.143**	0.241	0.150	0.251	0.145	0.244	0.315	0.389	0.235	0.322	0.297	0.367	0.484	0.518	0.697	0.638	0.639	0.609	1.265	0.919	1.414	0.855
	192	**0.155**	**0.240**	0.158	0.241	0.159	0.255	0.163	0.263	0.163	0.260	0.318	0.396	0.247	0.341	0.308	0.375	0.501	0.531	0.718	0.648	0.772	0.678	1.298	0.939	1.240	0.919
	336	**0.174**	**0.273**	0.178	0.277	0.179	0.274	0.175	0.278	0.183	0.281	0.340	0.415	0.267	0.356	0.354	0.411	0.574	0.578	0.758	0.667	0.901	0.745	1.302	0.942	1.253	0.921
	720	0.220	**0.308**	0.224	0.312	0.233	0.323	**0.219**	0.311	0.233	0.323	0.635	0.613	0.318	0.394	0.426	0.466	0.952	0.786	1.028	0.788	1.200	0.871	1.259	0.919	1.249	0.921
	Avg	**0.173**	**0.265**	0.179	0.268	0.178	0.273	0.176	0.275	0.181	0.277	0.402	0.453	0.266	0.353	0.346	0.404	0.627	0.603	0.800	0.685	0.878	0.725	1.281	0.929	1.289	0.904
Traffic	96	0.410	0.287	0.414	0.291	0.419	0.298	0.427	0.304	**0.404**	**0.286**	0.854	0.492	0.670	0.421	0.795	0.481	1.468	0.821	1.643	0.855	1.157	0.636	1.557	0.821	1.586	0.841
	192	0.415	**0.290**	0.419	0.291	0.434	0.305	0.447	0.315	**0.412**	0.294	0.894	0.517	0.653	0.405	0.837	0.503	1.509	0.838	1.856	0.928	1.688	0.848	1.596	0.834	1.602	0.844
	336	0.439	0.316	**0.437**	0.314	0.449	0.313	0.478	0.333	0.439	**0.310**	0.853	0.471	0.707	0.445	0.867	0.523	1.602	0.860	2.080	0.999	1.826	0.903	1.621	0.841	1.668	0.868
	720	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Avg	0.421	0.298	0.423	0.298	0.434	0.305	0.450	0.317	**0.418**	**0.296**	0.867	0.493	0.676	0.423	0.833	0.502	1.526	0.839	1.859	0.927	1.557	0.795	1.591	0.832	1.618	0.851
1st Count		**39**		10		2		8		6		0		2		0		0		0		0		0		0	

Table 15: Full zero-shot learning results on ETT datasets. A lower value indicates better performance. **Red**: the best, Blue: the second best, *Green*: the third best.

Methods		RDTU		Time-LLM		LLMTime		GPT4TS		DLinear		PatchTST		TimesNet		Autoformer	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1 → ETTh2	96	**0.272**	**0.329**	0.279	0.337	0.510	0.576	0.335	0.374	0.347	0.400	*0.304*	*0.350*	0.358	0.387	0.469	0.486
	192	**0.343**	**0.368**	0.351	0.374	0.523	0.586	0.412	0.417	0.447	0.460	*0.386*	*0.400*	0.427	0.429	0.634	0.567
	336	**0.378**	**0.407**	0.388	0.415	0.640	0.637	0.441	0.444	0.515	0.505	*0.414*	*0.428*	0.449	0.451	0.655	0.588
	720	**0.384**	**0.413**	0.391	0.420	2.296	1.034	0.438	0.452	0.665	0.589	*0.419*	*0.443*	0.448	0.458	0.570	0.549
	Avg	**0.344**	**0.379**	0.353	0.387	0.992	0.708	0.406	0.422	0.493	0.488	*0.380*	*0.405*	0.421	0.431	0.582	0.548
ETTh1 → ETTm2	96	**0.178**	**0.282**	0.189	0.293	0.646	0.563	0.236	0.315	0.255	0.357	*0.215*	*0.304*	0.239	0.313	0.352	0.432
	192	**0.231**	**0.307**	0.237	0.312	0.934	0.654	0.287	0.342	0.338	0.413	*0.275*	*0.339*	0.291	0.342	0.413	0.460
	336	**0.287**	**0.359**	0.291	0.365	1.157	0.728	0.341	0.374	0.425	0.465	*0.334*	*0.373*	0.342	0.371	0.465	0.489
	720	**0.367**	**0.383**	0.372	0.390	4.730	1.531	0.435	*0.422*	0.640	0.573	*0.431*	0.424	0.434	0.419	0.599	0.551
	Avg	**0.266**	**0.333**	0.273	0.340	1.867	0.869	0.325	0.363	0.415	0.452	*0.314*	*0.360*	0.327	0.361	0.457	0.483
ETTh2 → ETTh1	96	**0.446**	**0.449**	0.450	0.452	1.130	0.777	0.732	0.577	0.689	0.555	*0.485*	*0.465*	0.848	0.601	0.693	0.569
	192	**0.462**	**0.459**	0.465	0.461	1.242	0.820	0.758	0.559	0.707	0.568	*0.565*	*0.509*	0.860	0.610	0.760	0.601
	336	0.503	**0.480**	**0.501**	0.482	1.328	0.864	0.759	0.578	0.710	0.577	*0.581*	*0.515*	0.867	0.626	0.781	0.619
	720	0.509	**0.495**	**0.501**	0.502	4.145	1.461	0.781	0.597	0.704	0.596	*0.628*	*0.561*	0.887	0.648	0.796	0.644
	Avg	0.480	**0.471**	**0.479**	0.474	1.961	0.981	0.757	0.578	0.703	0.574	*0.565*	*0.513*	0.865	0.621	0.757	0.608
ETTh2 → ETTm2	96	**0.169**	**0.269**	0.174	0.276	0.646	0.563	0.253	0.329	0.240	0.336	*0.226*	*0.309*	0.248	0.324	0.263	0.352
	192	**0.228**	**0.307**	0.233	0.315	0.934	0.654	0.293	0.346	0.295	0.369	*0.289*	*0.345*	0.296	0.352	0.326	0.389
	336	**0.288**	**0.330**	0.291	0.337	1.157	0.728	0.347	*0.376*	*0.345*	0.397	0.348	0.379	0.353	0.383	0.387	0.426
	720	**0.387**	**0.411**	0.392	0.417	4.730	1.531	0.446	0.429	*0.432*	0.442	0.439	0.427	0.471	0.446	0.487	0.478
	Avg	**0.268**	**0.329**	0.272	0.341	1.867	0.869	0.335	0.370	0.328	0.386	*0.325*	*0.365*	0.342	0.376	0.366	0.411
ETTm1 → ETTh2	96	**0.319**	**0.363**	0.321	0.369	0.510	0.576	*0.353*	0.392	0.365	0.415	0.354	*0.385*	0.377	0.407	0.435	0.470
	192	**0.383**	**0.406**	0.389	0.410	0.523	0.586	*0.443*	0.437	0.454	0.462	0.447	*0.434*	0.471	0.453	0.495	0.489
	336	**0.401**	**0.428**	0.408	0.433	0.640	0.637	*0.469*	*0.461*	0.496	0.494	0.481	0.463	0.472	0.484	0.470	0.472
	720	0.407	**0.433**	**0.406**	0.436	2.296	1.034	*0.466*	*0.468*	0.541	0.529	0.474	0.471	0.495	0.482	0.480	0.485
	Avg	**0.378**	**0.408**	0.381	0.412	0.992	0.708	*0.433*	0.439	0.464	0.475	0.439	*0.438*	0.457	0.454	0.470	0.479
ETTm1 → ETTm2	96	**0.165**	**0.254**	0.169	0.257	0.646	0.563	0.217	0.294	0.221	0.314	*0.195*	*0.271*	0.222	0.295	0.385	0.457
	192	**0.225**	**0.315**	0.227	0.318	0.934	0.654	0.277	0.327	0.286	0.359	*0.258*	*0.311*	0.288	0.337	0.433	0.469
	336	**0.282**	**0.334**	0.290	0.338	1.157	0.728	0.331	0.360	0.357	0.406	*0.317*	*0.348*	0.341	0.367	0.476	0.477
	720	**0.372**	**0.362**	0.375	0.367	4.730	1.531	0.429	0.413	0.476	0.476	*0.416*	*0.404*	0.436	0.418	0.582	0.535
	Avg	**0.261**	**0.316**	0.268	0.320	1.867	0.869	0.313	0.348	0.335	0.389	*0.296*	*0.334*	0.322	0.354	0.469	0.484
ETTm2 → ETTh2	96	**0.295**	**0.351**	0.298	0.356	0.510	0.576	0.360	0.401	0.333	0.391	*0.327*	*0.367*	0.360	0.401	0.353	0.393
	192	**0.356**	**0.392**	0.359	0.397	0.523	0.586	0.434	0.437	0.441	0.456	*0.411*	*0.418*	0.434	0.437	0.432	0.437
	336	**0.363**	**0.407**	0.367	0.412	0.640	0.637	0.460	0.459	0.505	0.503	*0.439*	*0.447*	0.460	0.459	0.452	0.459
	720	**0.388**	**0.429**	0.393	0.434	2.296	1.034	0.485	0.477	0.543	0.534	0.459	0.470	0.485	0.477	*0.453*	*0.467*
	Avg	**0.351**	**0.395**	0.354	0.400	0.992	0.708	0.435	0.443	0.455	0.471	*0.409*	*0.425*	0.435	0.443	0.423	0.439
ETTm2 → ETTm1	96	**0.356**	**0.394**	0.359	0.397	1.179	0.781	0.747	0.558	0.570	0.490	*0.491*	*0.437*	0.747	0.558	0.735	0.576
	192	**0.386**	**0.414**	0.390	0.420	1.327	0.846	0.781	0.560	0.590	0.506	*0.530*	*0.470*	0.781	0.560	0.753	0.586
	336	**0.419**	**0.440**	0.421	0.445	1.478	0.902	0.778	0.578	0.706	0.567	*0.565*	*0.497*	0.778	0.578	0.750	0.593
	720	**0.485**	**0.487**	0.487	0.488	3.749	1.408	0.769	0.573	0.731	0.584	*0.686*	*0.565*	0.769	0.573	0.782	0.609
	Avg	**0.412**	**0.434**	0.414	0.438	1.933	0.984	0.769	0.567	0.649	0.537	*0.568*	*0.492*	0.769	0.567	0.755	0.591

Table 16: Zero-shot length generalization of RDTU. Models trained only up to $H = 336$ under the 5% data regime are directly evaluated on $H = 720$ without additional training, and are compared with RDTU trained on $H = 720$ using 10% data.

Dataset	**Metric**	**RDTU Trained on $H = 336$ w/ 5% Data, Eval on $H = 720$**	**RDTU Trained on $H = 720$ w/ 10% Data**
ETTh1	MSE / MAE	0.705 / 0.598	0.694 / 0.587
ETTh2	MSE / MAE	0.442 / 0.456	0.425 / 0.443
Traffic	MSE / MAE	0.478 / 0.337	0.463 / 0.325

Table 17: Cross-domain zero-shot generalization results. All zero-shot models are trained on ETTh1 and directly evaluated on target datasets from different domains without target-domain training. Lower MSE and MAE indicate better performance.

Target Dataset	Metric	Supervised RDTU	Zero-Shot RDTU	Zero-Shot Time-LLM	Zero-Shot GPT4TS
Electricity	MSE / MAE	0.156 / 0.246	**0.182 / 0.270**	0.254 / 0.312	0.328 / 0.385
Weather	MSE / MAE	0.221 / 0.251	**0.245 / 0.278**	0.386 / 0.395	0.462 / 0.431
Traffic	MSE / MAE	0.389 / 0.255	**0.421 / 0.295**	0.652 / 0.415	0.735 / 0.484
Illness	MSE / MAE	1.559 / 0.812	**1.645 / 0.910**	2.642 / 1.753	3.215 / 2.058

Table 18: Ablation study on RDFR training data construction on ETTh1 with prediction horizon $H = 96$. We compare RDFR trained on the full training set and the proposed Dual-Stream selected subset.

RDFR Training Data	**MSE** $\downarrow$	**MAE** $\downarrow$	**Training Time**
Full Dataset	0.358	0.388	> 3h
Dual-Stream Selected (Ours)	**0.351**	**0.382**	**0.3h**

Table 1: Average long-term forecasting results on the benchmark datasets. We followed the same setting in DMMV [Shen et al., 2025]. We set the forecasting horizons $H \in \{24, 36, 48, 60\}$ for Illness and $\{96, 192, 336, 720\}$ for the others. **Red**: the best, Blue: the second best.

View	Language						Multi-Modal				Visual		Numerical			
Methods	RDTU (**Ours**)		GPT4TS		Time-LLM		DMMV-A		Time-VLM		VisionTS		PatchTST		TimesNet	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	0.399	**0.413**	0.418	0.421	0.418	0.432	**0.395**	0.414	0.405	0.420	0.407	0.419	0.413	0.431	0.458	0.450
ETTh2	0.337	**0.377**	0.354	0.389	0.361	0.396	0.337	0.388	0.341	0.391	0.351	0.386	**0.330**	0.379	0.414	0.427
ETTm1	**0.340**	**0.367**	0.363	0.378	0.356	0.377	**0.340**	0.371	0.351	0.376	0.344	0.373	0.351	0.381	0.400	0.406
ETTm2	0.250	**0.309**	0.254	0.311	0.261	0.316	0.256	0.317	**0.248**	0.311	0.267	0.327	0.255	0.315	0.291	0.333
Illness	1.559	0.812	1.871	0.852	2.018	0.894	**1.407**	**0.771**	–	–	1.482	0.796	1.443	0.798	2.139	0.931
Electricity	**0.156**	**0.246**	0.170	0.263	0.165	0.259	0.158	0.248	0.172	0.272	0.159	0.250	0.162	0.253	0.193	0.295
Weather	0.221	**0.251**	0.227	0.255	0.244	0.270	**0.217**	0.256	0.224	0.263	0.225	0.258	0.226	0.264	0.259	0.287
Traffic	0.389	**0.255**	0.421	0.274	0.422	0.281	0.389	0.257	0.419	0.304	**0.386**	0.256	0.391	0.264	0.620	0.336
# Wins	9		0		0		5		1		1		1		0	

Table 2: Few-shot and zero-shot forecasting results, averaged over prediction horizons. **Bold red**: the best; Blue: the second best.

(a) Few-shot 10% training data

Dataset	RDTU	Time-LLM	GPT4TS
ETTh1	**0.551**	0.556	0.590
ETTh2	**0.367**	0.370	0.397
ETTm1	**0.402**	0.404	0.464
ETTm2	**0.274**	0.277	0.293
Weather	**0.232**	0.234	0.238
Electricity	**0.173**	0.175	0.176
Traffic	**0.426**	0.429	0.440

(b) Zero-shot ETT transfer

Source → Target	RDTU	Time-LLM	LLMTime
ETTh1 → ETTh2	**0.344**	0.353	0.992
ETTh1 → ETTm2	**0.266**	0.273	1.867
ETTh2 → ETTh1	0.480	**0.479**	1.961
ETTh2 → ETTm2	**0.268**	0.272	1.867
ETTm1 → ETTh2	**0.378**	0.381	0.992
ETTm1 → ETTm2	**0.261**	0.268	1.867
ETTm2 → ETTh2	**0.351**	0.354	0.992

Table 3: Ablation study of different reward components in the RDFR training stage. Improvements are computed relative to STIT model.

Method	**MSE** ↓	**MAE** ↓
Qwen2.5-Instruct-STIT	0.403	0.428
+ R_{len}	0.401 (-0.50%)	0.425 (-0.70%)
+ R_{fmt}	0.395 (-1.99%)	0.419 (-2.10%)
+ R_{acc}	0.367 (-8.93%)	0.398 (-7.01%)
+ $R_{len} + R_{fmt}$	0.388 (-3.72%)	0.412 (-3.74%)
RDTU (R_{total})	**0.351 (-12.90%)**	**0.382 (-10.75%)**